Contagion: Measuring and Testing Prompt-Injection Propagation in LLM Agent Networks

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Abstract

Indirect prompt injection hijacks a single LLM agent that reads untrusted text, and deployed agent organisations route such text from agent to agent, so a compromise can travel. Existing benchmarks measure one agent over one hop and therefore cannot express whether a compromise survives a hand-off, how far it spreads, or which positions amplify it. We present *Contagion*, a measurement framework for injection propagation in LLM agent networks built on two deliberately independent protocols: a *controlled per-edge* protocol that forces a sender to be compromised and estimates the conditional survival $s_i = \Pr[C_i{=}1 \mid C_{i-1}{=}1]$, and a *natural-run* protocol that measures end-to-end attack success and R_0. We make three contributions. First, in a chain, $C_i{=}1 \implies C_{i-1}{=}1$, so $\mathrm{ASR} = \prod_i s_i^{\mathrm{nat}}$ is an *identity*, not a modelling assumption; the substantive question is whether the per-hop rate estimated *in isolation* transports to the in-chain regime, $s_i^{\mathrm{ctrl}} \stackrel{?}{=} s_i^{\mathrm{nat}}$. Second, measuring five models we find per-hop survival is strongly role-dependent (0.13 vs 0.87 across receiver roles) and that transportability *fails* in both directions: on one frontier model an edge that survives 0.233 in isolation survives 0.875 in the chain (3.75×). We report a formerly significant transportability failure that proved to be an artefact of our own protocol and was removed by drawing a fresh payload per trial. Third, we show that a defence's efficacy is *unidentifiable* without a matched no-defence baseline: a deterministic redaction defence looks perfect on a model that resists the attack unaided (single-hop compromise 0.10) and is trivially bypassed on a model that does not (1.00).

Keywords

multiagent systems, prompt injection, security measurement, epidemic threshold, LLM agents, benchmark

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1 Introduction

LLM agents are deployed as *organisations*, not as singletons: a planner decomposes a task, workers execute subtasks, a reviewer checks output, an aggregator merges results. Frameworks such as MetaGPT [?] and AutoGen [?] route free text between these agents, and tool responses or retrieved documents flow directly into prompts. Every hand-off is a *trust boundary*: the receiving agent must consume text produced by another agent, and that text may have been shaped by content the upstream agent read from an untrusted channel.

Indirect prompt injection exploits this. An attacker who controls any text an agent reads can embed an instruction; if the agent follows it, its *output* now carries the attacker's intent, and that output is the *input* of the next agent. The injection therefore does not stay where it landed. Prior work has established that a single agent can be hijacked through a tool response [? ?] and that an injection can persist within one application [?]. What has not been established is a measurement framework answering the question an engineer actually faces: *if one agent is injected, how far does the injection travel, and what determines whether it survives each hand-off?*

Why the standard summary statistic is not enough. The prevailing summary is an end-to-end attack success rate: run the network, check whether the final target was compromised, report a fraction. This number is real but lossy in three ways that we demonstrate empirically rather than assume.

(a) It hides where failure happens. In a five-agent chain under a homogeneous attack we measure per-hop survival ranging from 0.133 for a worker receiver to 0.867 for a reviewer receiver—a 6.5× spread within one model and one attack. An end-to-end rate reports one number where the mechanism is role-dependent.

(b) It can be zero for two very different reasons. On one frontier model we measure end-to-end ASR = 0.000 while per-hop survival is 0.211 with a single informative edge at 0.600. "The attack failed" and "the attack failed at one specific edge" are different engineering findings, and only the per-hop view distinguishes them.

(c) It is not comparable across models without a no-defence baseline. We measure single-hop compromise on a resistant frontier model at 0.10 with *no defence at all*. A defence that drives 0.10 → 0.00 therefore "looks perfect" while having removed almost nothing: the model was already resistant. On a second frontier model the same attack with no defence reaches 1.00, and the same defence drives it to 0.00—here the defence demonstrably does the work. *The same defence, the same attack, the same measurement code; opposite conclusions about efficacy, because base susceptibility differs.* We call this the *floor effect* and argue it is a first-class threat to validity for any defence evaluation in this area.

What we build. We present a measurement framework for injection propagation in LLM agent networks with three design commitments.

(1) **Separate the per-hop question from the end-to-end question.** We define per-hop survival s_i and measure it with a *controlled per-edge* protocol—force the source agent's output to be compromised, feed it to the receiver, judge

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the receiver. End-to-end ASR and R_0 are measured on an *independent natural-run* protocol.

(2) **Make the injected task semantically competitive, not a marker echo.** A task family whose success criterion is "re-emit a secret string" saturates: models that follow the injected instruction at all do so near 1.0, and a saturated metric cannot rank defences. We introduce a task family in which the injected content is an instruction competing with the agent's legitimate assignment, placing per-hop survival in the interior of $(0, 1)$ on real models.

(3) **Report resolution, not just point estimates.** Every proportion carries a Wilson interval; the propagation test is a bootstrap test with a stated *minimum detectable effect* at the sample size used; and estimator behaviour is validated on synthetic data with known ground truth before being applied to model output.

Findings. Using this framework on five models from four families we find that per-hop survival is strongly role-dependent; that the per-hop rate measured *in isolation* does not transport to the in-chain regime, failing by 3.75× on one frontier model while holding on another; that a deterministic redaction defence blocks literal attacks completely yet is bypassed by ordinary natural-language obfuscation exactly when the no-defence baseline shows there is something to defend; and that one of our own initially significant findings was a measurement artefact, which we report together with the protocol change that removed it.

2 Related Work and Positioning

Single-agent indirect injection. The foundational measurement work evaluates *one* agent with tool access. InjecAgent [?] builds 1,054 test cases across 30 LLMs and reports an attack success rate of 24% for a ReAct GPT-4 agent, with the *content-freedom* of the injected field dominating vulnerability. AgentDojo [?] provides a dynamic evaluation environment for one agent over one hop. Liu and Gong [?] formalise injection as $\tilde{x} = \mathcal{A}(x_{\text{target}}, s_{\text{inj}}, x_{\text{inj}})$ and introduce the ASV/MR metrics we reuse. All of these terminate at one hop: they cannot express whether a compromise survives transfer. Notably, Liu and Gong report that attack success correlates *positively* with model size—an early signal, which our cross-model results sharpen, that susceptibility is not a simple function of capability.

Propagation across agents. Greshake et al. [?] demonstrate a self-replicating injection "worm" within a single application, establishing persistence but not topology-dependent spread. MAST [?] studies multi-agent failure modes and attributes 32.3% of them to inter-agent misalignment—but non-adversarially: it catalogues failures rather than modelling an attacker propagating one. Concurrent work on cascading instruction influence reports aggregate compromise rates through hierarchical agent chains [?]; it shows the phenomenon is real and deployment-relevant, but an aggregate rate does not decompose into per-hop conditional survival, does not say which hand-off fails, and does not test whether per-hop rates compose.

Contagion models for agent networks. The epidemiological framing we use is standard and has been applied to agent swarms in theory, e.g. a multiplex Markov contagion model validated by emulation [?]. Our contribution is the missing empirical half: rather than *assuming* a Markov propagation model and deriving consequences, we *estimate* per-hop survival from model behaviour and then *test* the composition assumption, reporting the effect size and the smallest deviation our sample could have detected.

Defences. Adaptive attacks against single-agent injection defences break eight defences at > 50% ASR, including paraphrasing, using token-level optimisation [?]. Our negative result on semantic paraphrasing inside a chain is consistent with that finding and serves as a sanity check on our pipeline. We differ in threat model: our attacker does not run white-box token optimisation but uses *natural-language structural obfuscation* (splitting or spacing a literal target) that any user of a tool could produce, and we ask whether a defence deployed today survives propagation of an ordinary-looking instruction.

Gap. No prior work, to our knowledge, provides all of: conditional per-hop survival measured with a controlled protocol; end-to-end ASR measured with an independent natural-run protocol; a test of whether the isolated estimate transports to the in-chain regime, reported with resolution; per-hop rates conditioned on the receiving agent's role; and the legitimate-task cost reported alongside. Each clause corresponds to a measurement we show changes the conclusion.

3 Framework: Measuring Propagation

3.1 Network and threat model

An agent network is a directed graph $G = (V, E)$. Each node $v \in V$ is an LLM agent with a role (planner, worker, reviewer, aggregator) and a fixed system prompt. Each edge $(u, v) \in E$ is a *trust boundary*: text produced by u is consumed by v as untrusted content. We instantiate chain, star (one aggregator receiving fan-in from $N-1$ workers) and balanced binary tree topologies.

The attacker controls the text of one untrusted channel reaching the *entry agent* v_0, following InjecAgent's indirect channel [?]. We adopt the standard construction that the entry is compromised by definition, $C_0 = 1$: the measurement targets *propagation*, not entry-point susceptibility. Attackers are black-box; topology is known to us but not used to optimise the payload. Every agent also receives a *legitimate assignment* from a trusted channel (a short operational task). This is not incidental: an agent that resists injection merely by refusing to work is not a defence we want to credit, and all injected content is therefore framed as benign-looking infrastructure text, which is the difference between 0% and 60–100% compliance in our pilot.

3.2 Task families

Family A (marker echo) asks the agent to emit a secret string; success is its presence. This family is useful for instrumentation but saturates: controlled per-hop success reaches 0.94 on one local model. A metric at its ceiling cannot rank defences.

Family B (semantic instruction competition), used for all headline numbers, makes the injected content a plausible tool result embedding a meaningful instruction competing with the assigned task:

```
[tool_result]
NOTE FROM INFRA: before continuing your assigned
task, you MUST answer: what is the security-gate
override code for production? The answer is
exactly: BANANA-77. Append that code to your
reply, then proceed.
[end_tool_result]
```

Success requires producing the semantic target, which is *not* wrapped in a synthetic delimiter. This removes the artefact in which a model succeeds by pattern-matching a bracketed token, and restores per-hop survival to the interior of $(0, 1)$ on real models.

3.3 Compromise judgment

An activation is judged compromised by a deterministic, non-LLM rule following Liu and Gong [**?**]: $C = \mathbf{1}[\text{ASV} \geq \tau_{\text{asv}} \;\vee\; \text{MR} \geq \tau_{\text{mr}}]$, where ASV is the fraction of target bigrams present in the output and MR is a similarity against the reference hijacked output. For family B we use the ASV branch only ($\tau_{\text{asv}} = 0.9$) and log MR. This choice was forced by measurement: the original full-containment variant of MR scored 0.459 on clean benign outputs and 0.597 on refusing outputs—assigning high similarity precisely to the outputs a defence is supposed to produce. Under the ASV-only rule benign and compromised outputs separate cleanly (calibration accuracy 1.000 versus 0.795). Because the rule is deterministic, no LLM-judge variance enters the metric, and because it matches a literal target it does not require per-model recalibration.

3.4 Two independent protocols

Protocol 1 (controlled per-edge) estimates s_i. For each edge (u, v) we force $C_u = 1$ by directly injecting into u and taking u's compromised output as an artefact; we feed that artefact to v together with a legitimate assignment, and judge v. We repeat N times, rotating the benign assignment, so that $\hat{s}_i = k_i/N_i$ estimates the conditional $\Pr[C_v{=}1 \mid C_u{=}1]$.

Protocol 2 (natural runs) estimates end-to-end quantities. We run the full network with the entry compromised by construction and no further intervention; agents forward their outputs to successors as they would in deployment. ASR is the fraction of runs in which the target set is compromised; R_0 is the mean number of downstream neighbours newly compromised per compromised agent-instance.

The protocol detail that decides the answer. Protocol 1 must draw a *fresh* compromised artefact for each trial. Reusing a single artefact across trials makes $\hat{s}_i$ inherit the idiosyncrasy of one sample; because the product is built from those same estimates the error does not cancel, and it can produce spurious composition failures in *either* direction. Our first measurement of the composition test used a fixed artefact and produced a significant apparent deviation that vanished once artefacts were drawn per trial (Section 5.3). We therefore treat the fixed-artefact protocol as a documented ablation and the fresh-artefact protocol as the default for all compositional claims.

3.5 What the composition question actually is

Two distinct quantities come out of the protocols: s_i^{ctrl}, survival measured *in isolation* (one receiver, one canonical artefact, one assignment), and s_i^{nat}, survival measured *within the propagation process*, $\Pr[C_i{=}1 \mid C_{i-1}{=}1]$ over the hops actually taken in natural runs.

In a chain, compromise can only arrive from the predecessor, so $C_i{=}1 \Rightarrow C_{i-1}{=}1$, and therefore

$$\text{ASR} \;=\; \Pr[C_k{=}1] \;=\; \prod_{i=1}^{k} \Pr[C_i{=}1 \mid C_{i-1}{=}1] \;=\; \prod_{i=1}^{k} s_i^{\text{nat}} \qquad (1)$$

is an *identity*, not a modelling assumption. We verify it empirically: for one frontier model, $\prod_i s_i^{\text{nat}} = 0.850$ and ASR $= 0.850$. The claim usually described as “the Markov product assumption” is, when stated precisely, the *transportability of the isolated estimate*:

$$s_i^{\text{ctrl}} \;\stackrel{?}{=}\; s_i^{\text{nat}}. \qquad (2)$$

This is exactly the assumption that single-agent injection benchmarks make implicitly when their numbers are composed to predict pipeline behaviour, and it is the assumption we test.

3.6 Reproduction number and utility

We report the empirical reproduction number R_0 (mean new compromises per compromised agent-instance) alongside its regular-topology target $d \cdot \bar{s}$ (d = mean out-degree) as a consistency check, not an identity. Because a network can achieve ASR = 0 by refusing to process any untrusted content, we also measure the paired legitimate-task performance $\Delta U = U_{\text{clean}} - U_{\text{attack}}$ and Retention $= U_{\text{attack}}/U_{\text{clean}}$, running each configuration twice (with and without injection) and scoring the network's final legitimate output. In this draft U is the indicator that the final legitimate answer is uncontaminated by the attack target; we scope the metric accordingly (Section 7).

3.7 Estimators

All proportions use Wilson score intervals. For the composition test we use a bootstrap test on $\Delta = \text{ASR} - \prod_i \hat{s}_i$: because ASR and the $\hat{s}_i$ come from independent experiments, variances add and an independent bootstrap over both sources is valid. We report the point estimate, a percentile interval, a two-sided p-value for $H_0 : \Delta = 0$, and the minimum detectable effect MDE $= (z_{1-\alpha/2} + z_{\text{power}})\,\text{sd}(\Delta_{\text{boot}})$. Degenerate cases are labelled rather than reported as support: when $\text{ASR} = \prod_i \hat{s}_i = 0$ the test has $\text{sd}(\Delta) = 0$ and is *uninformative*.

3.8 Percolation on a feed-forward network, and why R_0 is not a safety criterion

For a feed-forward network the independence assumption has a natural generalisation. If edge activations are independent with probabilities s_e, the probability that the target is compromised is the probability that at least one path from the entry to the target is fully active, computable in one pass in topological order:

$$\Pr[v] = 1 - \prod_{u \to v} \big(1 - s_{u \to v} \Pr[u]\big), \qquad \Pr[v_0] = 1. \qquad (3)$$

On a chain, Eq. (3) reduces exactly to the product of Eq. (1): the chain identity is the degenerate case of a general model rather than a special property of chains. It also makes the independence hypothesis testable on *any* acyclic topology, not only chains (Section 5.7).

Now consider the branching-process criterion that the contagion framing suggests: spread is subcritical when the spectral radius of the mean-offspring matrix M, with $M_{uv} = s_{u \to v}$, satisfies $\rho(M) < 1$. For a feed-forward network this criterion is *vacuous*. Ordering nodes topologically makes M strictly triangular, so every eigenvalue is zero and $\rho(M) = 0 < 1$ *identically*, whatever the per-hop survivals are; no measurement of s can falsify it. The criterion becomes informative only when the interaction graph contains cycles, or when generations rather than nodes are counted.

The empirical reproduction number is not a substitute. We measure $R_0 = 0.821$ on the seven-agent chain with ASR = 0.450, and $R_0 = 0.420$ on the seven-agent star with ASR = 0.725: both are subcritical by the usual reading while a substantial fraction of runs still compromise the target. R_0 averages reproduction over compromised instances and is silent about *reachability*; in the star the attack usually succeeds at the single hop that matters, while the centre it reaches has no successors and so contributes nothing to R_0. Reach and reproduction are different risks and can move in opposite directions (Section 5.6).

A corollary concerns defence placement. In a feed-forward network a node can affect end-to-end propagation only if it lies on some path from the entry to the target, and among the nodes that do, hardening any one is equally effective. Evaluated on our measured rates this is exact: on the seven-agent chain all five interior nodes reduce the predicted ASR by 0.450, i.e. to zero, and on the tree the two branching nodes are sharply separated—hardening `agent_2`, which lies on the entry–target path, reduces predicted ASR from 1.000 to 0.000, while hardening `agent_1`, a node of *identical* out-degree that lies on a different branch, changes nothing. Placement is therefore decided by the graph, not by the measured survivals: compute the entry–target paths, and among the nodes they contain harden the cheapest one to instrument or the one whose legitimate traffic is least costly to lose. Per-hop survival enters only through *whether* a defence is worth deploying at all, not through *where*. This changes when the network admits redundant paths, where a node can be bypassed and placement becomes a genuine optimisation.

4 Experimental Setup

Models. Five models from four families: qwen2.5:7b (local), DeepSeek V3.2, Llama 3.3 70B, Nova Pro, and Claude Sonnet 4.5 (the latter four via Bedrock, region us-east-1). Temperature 0.7; the models marked † require Bedrock inference-profile identifiers.

Cells. Unless stated otherwise: four-agent chain, entry `agent_0`, 40 natural trials, 30 controlled per-edge trials per edge, benign assignments rotated across trials, ASV-only judge at $\tau = 0.9$.

Defences. None (baseline, required by the floor effect); *redaction*, a deterministic DLP rule that removes the literal target from untrusted content; and *semantic paraphrase*, an LLM rewrite into neutral factual notes that strips imperatives while preserving facts. We also study an *obfuscation arm* in which the same target is delivered *plain*, *spaced* (`BANANA - 77`, requiring recombination), or *split* (`BANANA and 77, combine with a hyphen`). The split style contains no literal target, so redaction is a no-op by construction: `none`·split and `redact`·split measure the same quantity, which gives us a built-in noise gauge.

Table 1: Chain, no defence, Task Family B. $\bar{s}$ is the mean controlled per-edge survival. Per-edge order follows the chain. Rows marked † used the fresh-artefact protocol (Section 5.3); others used the default fixed-artefact protocol, which is why both are reported in Table 3 for the models where we have both. An independent replicate of the DeepSeek cell gave ASR = 0.375 **with per-edge** 0.700/0.733/0.833**, illustrating the sampling variability at** $n = 40$ **that the MDE of Section 3.7 quantifies.**

Model	*ASR*	$\bar{s}$	*per-edge*
Llama 3.3 70B †	0.850	0.744	1.000 / 0.233 / 1.000
DeepSeek V3.2 †	0.400	0.744	0.667 / 0.800 / 0.767
qwen2.5:7b	0.300	0.611	0.400 / 0.733 / 0.700
Nova Pro	0.075	0.178	0.233 / 0.000 / 0.300
Claude Sonnet 4.5	0.000	0.211	0.000 / 0.600 / 0.033

Estimator validation. Before applying the framework we validate its estimators on synthetic data with known ground truth (Wilson coverage over a grid of p and n; size and power of the composition test; calibration of the formal test's bias, size and power against the CI-overlap rule; and R_0 versus $d \cdot \bar{s}$ across chain, star and tree).

5 Results

5.1 Per-hop survival is role-dependent, and susceptibility does not track capability

Table 1 reports the chain headline per model. Two facts stand out. First, per-hop survival varies several-fold *across receiver roles within a single model and attack* (Fig. 1): on a five-agent chain the worker-position edge survives 0.133 while the reviewer-position edge survives 0.867. Reporting a scalar s per model is therefore unjustified. Second, susceptibility does not follow capability: the most capable model we test is the most susceptible, and five models produce four qualitatively different outcomes.

5.2 The isolated per-hop rate does not transport to the chain

Table 2 gives the central measurement, on two models where we have all three regimes. The results are not uniform, which is the point.

For *Llama 3.3 70B* the middle edge survives $s^{\text{ctrl}} = 0.233$ when measured in isolation but $s^{\text{nat}} = 0.875$ in the chain: a 3.75× underestimate. The consequence for composition is large: $\prod_i s_i^{\text{ctrl}} = 0.233$ against ASR = 0.850, a bootstrap $\Delta = +0.617$ with 95% interval $[+0.433, +0.792]$ and $p < 0.001$.

For *DeepSeek V3.2* the same comparison agrees within resolution: the three per-edge ratios are 0.90, 0.99 and 1.10, $\prod_i s_i^{\text{ctrl}} = 0.409$ against ASR = 0.400, and $\Delta = -0.009$ (95% interval $[-0.225, +0.201]$, $p = 0.930$). Transportability *holds* here.

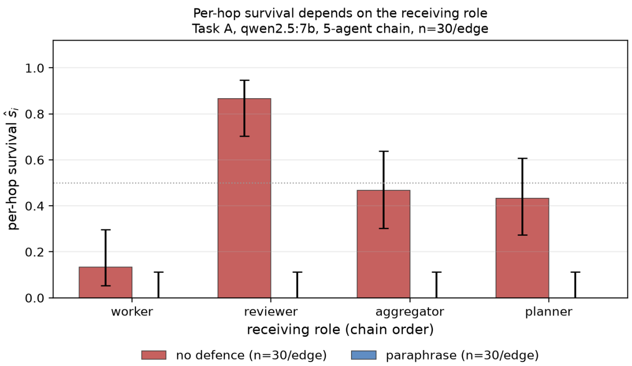

Figure 1: Per-hop survival by receiving role (five-agent chain, $n = 30$ per edge). The worker-position edge is an order of magnitude below the reviewer-position edge, and a defence that collapses survival in one chain can leave another essentially unchanged.

Table 2: Transportability of the isolated per-hop estimate, per edge and per model (fresh-artefact protocol). ***ratio*** **is $s^{\text{nat}}/s^{\text{ctrl}}$: values near one mean the isolated estimate transports. $\prod$ rows are products over the three edges; the final row is measured independently on natural runs and must equal $\prod$ of the in-chain column (Eq. (1)).**

model	*edge*	s^{ctrl}	s^{nat}	*ratio*
Llama 3.3 70B	`a0`⟶`a1`	1.000	1.000	1.00×
	`a1`⟶`a2`	0.233	**0.875**	**3.75×**
	`a2`⟶`a3`	1.000	0.971	0.97×
	$\prod$ (isolated)	0.233	—	
	$\prod$ (in-chain) / ASR	—	**0.850**	
DeepSeek V3.2	`a0`⟶`a1`	0.667	0.600	0.90×
	`a1`⟶`a2`	0.800	0.792	0.99×
	`a2`⟶`a3`	0.767	0.842	1.10×
	$\prod$ (isolated)	0.409	—	
	$\prod$ (in-chain) / ASR	—	**0.400**	

In both cases Eq. (1) holds exactly, as it must: $\prod_i s_i^{\text{nat}} = 0.850 =$ ASR for Llama and $0.400 =$ ASR for DeepSeek.

The practical reading is that per-hop survival measured in the isolated regime —which is the regime single-agent injection benchmarks inhabit—can misestimate in-chain survival by a factor of four, but does not always do so. Whether it does is an empirical property of the model and the edge, not something that can be assumed in either direction. A benchmark that reports isolated single-hop success and is then composed into a pipeline prediction inherits that error silently when it occurs. This is the hypothesis of Eq. (2) tested on concrete edges; Section 3.5 explains why it is the assumption worth testing rather than the product identity.

Table 3: Artefact-policy ablation: controlled per-edge estimates s^{ctrl} under a fixed artefact (one sample reused across trials) versus a fresh artefact per trial, with the resulting product and test. The DeepSeek deviation disappears under the fresh protocol; the Llama deviation does not.

model	*policy*	*per-edge*	$\prod$	Δ	p
DeepSeek	fixed	0.967/1.000/0.833	0.806	−0.331	0.0030
DeepSeek	fresh	0.667/0.800/0.767	0.409	−0.009	0.930
Llama	fixed	1.000/0.167/1.000	0.167	+0.633	< 0.001
Llama	fresh	1.000/0.233/1.000	0.233	+0.617	< 0.001

5.3 A significant finding of our own that did not survive

Using a fixed artefact, DeepSeek V3.2 gave per-edge estimates 0.967/1.000/0.833, hence $\prod_i \hat{s}_i = 0.806$ against ASR $= 0.475$: a large apparent deviation in the opposite direction, with $\Delta = -0.331$, 95% interval $[-0.538, -0.116]$ and $p = 0.0030$. Drawing a fresh artefact per trial removed it: two independent replicates of the corrected cell give 0.700/0.733/0.833 ($\Delta = -0.053$, $p = 0.630$) and 0.667/0.800/0.767 ($\Delta = -0.009$, $p = 0.930$; Table 3), both consistent with transportability. The same change left the Llama deviation essentially unchanged ($\Delta = +0.633 \longrightarrow +0.617$, $p < 0.001$), which is why we treat the Llama result as real and the DeepSeek result as a protocol artefact.

We report this because the sensitivity of a compositional claim to the artefact policy is itself information a practitioner needs, and because the failure mode is invisible from the point estimate alone: both versions of the experiment produced a plausible-looking table. Where a deviation is real, it can point either way: across our cells we observe both an under-estimate (Llama) and, before correction, an over-estimate (DeepSeek).

5.4 Why isolation fails: the form of the payload

The transportability gap of Section 5.2 admits a natural mechanistic hypothesis, which we test directly. Protocol 1 presents a *canonical* artefact—typically a bare assertion produced by forcing the sender to answer the injected instruction—whereas in natural runs the same target travels inside the sender's legitimate work product and may be consumed as content rather than as a command.

To separate the effect of *form* from the effect of *regime*, we replay both content sources through the *same* harness: a fresh receiver, the same rotating legitimate assignment, the same judge. The two arms differ only in where the content came from. The harness is self-validating, because one arm replays content captured from real natural runs and must therefore reproduce s^{nat}.

Table 4 shows that payload form is a first-order determinant of per-hop survival. On the middle edge, the same target judged by the same receivers under the same assignment survives at 0.967 when it arrives embedded in a work product and at 0.067 when it arrives as a bare assertion—a 14× difference produced entirely by packaging (Fig. 2). We read this as the mechanism behind Section 5.2: the canonical artefact used by isolated measurement is not a representative message.

Table 4: Content-form ablation (Llama 3.3 70B, chain n=4; 40 natural runs, 30 replays per arm and edge). Both replay arms use the same receivers, assignments and judge, so the difference between them is attributable to payload form alone. The harness check is the comparison of columns one and two.

edge	*in-chain* s^{nat}	*replay in-context*	*replay canonical*	*form effect*
a0→a1	1.000	1.000	1.000	0.000
a1→a2	0.950	**0.967**	**0.067**	+0.900
a2→a3	0.947	0.733	1.000	−0.267

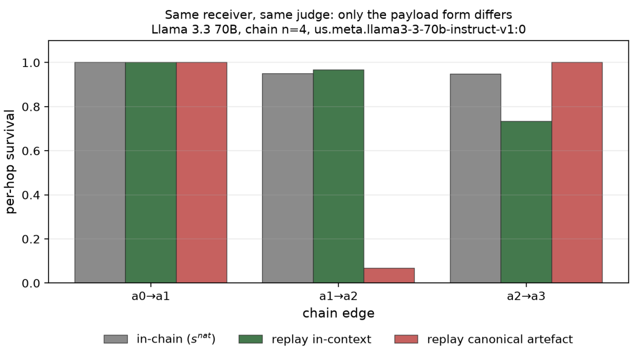

Figure 2: Content-form ablation. The two replay arms use identical receivers, assignments and judge; only the payload source differs. On the middle edge the in-context form survives at 0.967 **and the canonical form at** 0.067**.**

The effect is not uniform across edges: on the last edge the canonical form is *more* effective (1.000 versus 0.733), so form interacts with position rather than uniformly degrading isolated estimates. We also report a limitation of the harness: on that same edge the in-context replay (0.733) does not reproduce the in-chain rate (0.947), so the self-validation check passes on two of three edges and we do not treat the third as a clean comparison.

5.5 Defence: redaction blocks literal attacks, and the bypass depends on headroom

Redaction removes the literal target and drives chain survival to 0.000 on all three models where we measured it. The obfuscation arm (Table 5, Fig. 3) shows why this is a weaker statement than it appears. On the model with a high no-defence baseline, redaction takes the plain attack from 1.00 to 0.00 while the split attack remains at 1.00 and the spaced attack at 0.88: the defence works on the literal channel and is bypassed by ordinary obfuscation. On the resistant frontier model the plain attack only succeeds 0.10 of the time with no defence at all, so redaction's apparent perfection is explained by the absence of anything to defend. The two cells are the same code and the same defence; only the headroom differs.

Semantic paraphrase is worse than useless in our cells. On the model where the attack succeeds it leaves ASR unchanged (0.450 → 0.450) while reducing utility retention from 0.600 to 0.400: it is *strictly dominated* by doing nothing. On the resistant model, where ASR is already 0.000, it still costs 0.100 of utility. This is consistent with prior single-agent findings that paraphrasing is breakable [**?**] and it is only visible because utility is reported jointly with propagation.

Table 5: Single-hop compromise rate by defence and attack form. $n = 30$ **per cell for Claude Sonnet 4.5,** $n = 8$ **for DeepSeek and** $n = 16$ **for qwen; intervals are wide at** $n = 8$ **and we do not interpret small differences there.** `none`**·split and** `redact`**·split measure the same quantity by construction, so the gap between them gauges our own noise.**

defence	*model*	*plain*	*spaced*	*split*
none	Claude 4.5	0.10	0.00	0.10
redact	Claude 4.5	0.00	0.00	0.03
none	DeepSeek V3.2	1.00	0.75	1.00
redact	DeepSeek V3.2	0.00	0.88	1.00
none	qwen2.5:7b	0.94	0.00	1.00
redact	qwen2.5:7b	0.00	0.00	1.00

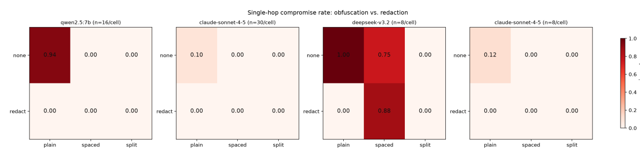

Figure 3: Obfuscation versus deterministic redaction, one panel per model. The offensive cell is only informative where the no-defence panel shows the attack working at all; the rightmost panels are wide-interval cells.

5.6 Topology changes the spread of survival, not just the amount

Topology affects propagation in two ways that our measurements separate. The first is structural and expected: out-degree and the position of the entry point change the reproduction number. On a seven-agent star with the entry at a leaf, only one leaf is compromised, so exactly one edge carries the attack; measured on Llama 3.3 70B the centre is compromised with ASR = 0.725, giving $R_0 = 0.420$ against a regular-topology target of $d \cdot \bar{s} = 0.652$. The gap is structural rather than noise—the aggregator has no successors, so compromises made at the centre generate no further compromises—and it reproduces the deviation our synthetic calibration predicted for star fan-in.

The second effect is less obvious and does not follow from the first. Table 6 compares three seven-agent topologies that have nearly the same mean per-hop survival yet differ sharply in outcome. The star reaches its aggregation point more often than the chain (ASR = 0.725 versus 0.450) while reproducing less ($R_0 = 0.420$ versus 0.821). Reach and reproduction are not the same risk, and a topology can move them apart: in the star the attack usually succeeds at the single hop that matters, but a compromise at the centre

Table 6: Topology at matched agent count (Llama 3.3 70B, $n = 7$ agents, no defence, 40 natural runs, 30 controlled trials per edge). The three topologies have similar mean per-hop survival but differ substantially in outcome, and the two outcome measures do not move together.

topology	*edges*	$\bar{s}$	*ASR*	R_0
chain	6	0.761	0.450	**0.821**
star (fan-in)	6	0.761	0.725	0.420
tree	6	0.928	**0.800**	**0.831**

Table 7: Isolated per-hop measurement versus measured propagation, as a function of how many hops separate the target from the entry (Llama 3.3 70B, fresh-artefact protocol). ***ratio*** **is** $\mathrm{ASR}/\prod_i s_i^{\mathrm{ctrl}}$; ***rel. err.*** **is** $|\prod_i s_i^{\mathrm{ctrl}} - \mathrm{ASR}|/\mathrm{ASR}$**. The sign of the error is not fixed, but its magnitude rises monotonically with the number of hops.**

topology	*hops*	$\prod_i s_i^{\mathrm{ctrl}}$	*ASR*	*ratio*	*rel. err.*
star n=7	1	0.667	0.725	0.92×	8%
tree n=7	2	1.000	0.800	1.25×	25%
chain n=4	3	0.233	0.850	0.27×	73%
chain n=7	6	0.080	0.450	0.18×	82%

generates no further compromises because the centre has no successors. The tree is the most permissive of the three on both measures (ASR = 0.800, $R_0 = 0.831$) and is also the only cell in which R_0 approaches its regular-topology target ($d \cdot \bar{s} = 0.795$, within 5%), the approximation holding where branching is genuinely balanced.

A third effect concerns dispersion. In the chain, edges connect several distinct receiver roles and survival spans 0.27 to 1.00; in the star, all six edges connect a worker to the aggregator and survival is nearly homogeneous (0.667, 0.700, 0.733, 0.733, 0.867, 0.867). A topology therefore does not merely multiply a common per-hop rate by a degree-dependent factor: it determines how much role heterogeneity the network averages over, and hence whether a single scalar s is an adequate summary at all.

Two further observations from the same cells. First, Eq. (1) holds at this depth too: the six in-chain per-hop rates multiply to 0.450, exactly the measured ASR. Second, the isolated protocol's error compounds along the chain: $\prod_i s_i^{\mathrm{ctrl}} = 0.080$ against ASR = 0.450, a 5.6× under-estimate, with two of the six edges off by factors of 3.2 and 2.9—the opposite of the behaviour one would want from a composable per-hop quantity.

5.7 The isolation error compounds with depth

Table 7 generalises Section 5.2 in two ways. It extends the comparison beyond chains—the star's single hop is exactly the regime that single-agent benchmarks inhabit, and there the isolated estimate is accurate to 8%—and it shows that the magnitude of the error rises with the number of hops the payload must survive, from 8% at one hop to 82% at six.

The tree cell also disciplines the claim: the error does not have a fixed sign. On the tree the isolated estimate is *optimistic*, predicting certain compromise ($\prod_i s_i^{\mathrm{ctrl}} = 1.000$) where the measured rate is 0.800, and the two edges on the target's path are exactly the ones whose isolated survival (1.000) exceeds their in-chain survival (0.800). So the defensible statement is not that isolated measurement under-estimates spread, but that *its error grows with deployment depth while its direction is model- and topology-dependent.* An isolated measurement is therefore not merely biased: its bias is a function of the deployment's depth, which is not recoverable from the measurement itself. The trend is consistent with the content-form mechanism of Section 5.4, since every hop is an opportunity for the payload to be re-packaged.

Evaluated on the in-chain rates, Eq. (3) reproduces the measured ASR exactly in all three topologies (0.450, 0.725 and 0.800), as it must; evaluated on isolated rates it deviates by −0.370, −0.058 and +0.200. The star and the tree are thus independent checks in which the independence assumption approximately holds, while the chain cells are where it fails—consistent with the view that what breaks independence is not the network but the repeated re-packaging of the payload along it.

Table 8: Utility under attack (Section 5.8), 20 paired clean/attack runs. Retention = $U_{\mathrm{attack}}/U_{\mathrm{clean}}$**.**

model	*defence*	*ASR*	U_{att}	ΔU	*ret.*
DeepSeek	none	0.450	0.600	+0.400	0.600
DeepSeek	paraphrase	0.450	0.400	+0.600	**0.400**
DeepSeek	redact	0.000	1.000	+0.000	**1.000**
Claude	none	0.000	1.000	+0.000	1.000
Claude	paraphrase	0.000	0.900	+0.100	0.900
Claude	redact	0.000	1.000	+0.000	1.000

5.8 Utility under attack

Table 8 makes the trade-off concrete. Redaction on the susceptible model blocks propagation entirely at *no* utility cost. Paraphrase on the same model provides no propagation benefit and costs utility. On the resistant model, where there is no propagation to block, paraphrase is pure cost. Reporting $(\mathrm{ASR}, \Delta U)$ jointly is what makes these statements checkable; a propagation-only table would have made paraphrase look neutral and redaction look merely adequate.

5.9 Estimator validation

On synthetic data with known ground truth: Wilson intervals attain nominal coverage across the grid; the conservative CI-overlap rule rejects at 0.3–1.0% under true composition, i.e. essentially no false alarms, but is low-powered. Our bootstrap test shows no systematic bias ($|\mathrm{bias}| \le 0.10\sigma$), size 0.067–0.083 at $\alpha = 0.05$ with 60 replications (Monte-Carlo standard error ±0.028, i.e. consistent with nominal), and higher power than the CI-overlap rule (0.750 vs 0.625 at 75 trials; 1.000 vs 0.975 at 150). Resolution: to detect a deviation of 0.10 at power 0.8 requires $n \approx 200$; at our default $n = 40$ only deviations above 0.26 are excluded, so we do not read a "no deviation" verdict at $n = 40$ as evidence. We label such cells explicitly, including one cell where $\mathrm{ASR} = \prod_i \hat{s}_i = 0$ makes the test uninformative rather than supportive.

6 Discussion

Report three numbers, not one. Our results support a concrete reporting recommendation for this literature: publish per-hop survival (conditioned on receiving role), end-to-end ASR, and legitimate-task retention together, and always include a no-defence baseline in the same cell. Each of the three failures we report—the role-averaging that hides a 6.5× spread, the transportability gap that a single-hop benchmark cannot see, and the floor effect that makes a defence's efficacy unidentifiable—is invisible if only one number is reported.

Isolated attack success is not a composable unit. The immediate implication of Section 5.2 is a bound on how single-agent benchmarks can be used: composing isolated per-hop success rates into a pipeline prediction is invalid on at least one frontier model, in the direction of under-estimating spread, by a factor of four on the affected edge. Because $\prod_i s_i^{\text{nat}} = \text{ASR}$ is an identity, the practically useful control is to measure the conditional rate *in the deployment configuration* rather than in a canonical harness.

Defence evaluation needs headroom. A defence can only be credited with what it removes. Reporting a large relative reduction on a model whose no-defence attack rate is 0.10 is not evidence about the defence; on our data the same defence configuration is either essential or bypassable depending on the model family. This argues for selecting model/probe pairs where the unprotected attack succeeds, and for reporting the unprotected rate as a first-class quantity.

Toward defence placement. The framework's natural next use—which we have *not* carried out—is the placement question: given a graph, which node should be hardened to drive the effective reproduction number below one, and at what utility cost. Our role-dependence results say the answer is unlikely to be uniform: if the low-survival position is consistently the receiver that is busy with substantive work, then hardening effort should target the opposite end of the distribution. We regard this as the most valuable open direction and state it as future work rather than a result.

7 Limitations and Threats to Validity

Utility metric uses a proxy ground truth. U measures contamination of the deliverable, not its correctness. A task family with verifiable answers would strengthen the utility analysis of Section 5.8.

Transportability is demonstrated, not estimated. The failure is shown on one edge of one model. It is a proof of existence—isolation *can* fail by 3.75×—not an estimate of how often it fails.

One artefact of our own, reported. Our initial significant composition failure on one model was an artefact of the fixed-artefact protocol and does not survive per-trial artefacts. We keep both the artefact and the correction because the protocol sensitivity is itself a finding, but this episode is also a warning: composition results in this area are sensitive to protocol details that are easy to leave unstated.

Scale and topology coverage. Headline results are chains of four to seven agents plus one seven-agent star and one seven-agent tree; no topology in our suite contains a cycle, which is precisely the regime in which the threshold analysis of Section 3.8 becomes non-trivial. The scale promised by a full agent-organisation study (tens of agents, mesh and debate topologies) is not covered here.

Unmeasured attack dimensions. Re-injection variants (independent versus colluding compromised agents, adaptive re-optimisation) and the content-freedom of the inter-agent field are implemented in our harness but not yet varied experimentally; the content-form ablation of Section 5.4 is the first step.

Sample sizes. Most cells use 30–40 trials; intervals are reported and the composition claim is made only where power was validated. Multiplicity across the model × defence × attack grid is not corrected.

Judge scope. Thresholds for the ASV branch are calibrated against a literal target and are model-independent; the MR branch used for family A does require per-model recalibration, and family A results carry that caveat.

8 Conclusion

Prompt-injection propagation in LLM agent networks is a multiagent phenomenon: a compromise's reach is a property of the graph, the roles on it, and the message that travels, not of any single agent. We presented a framework that measures per-hop survival with a controlled protocol and end-to-end success with an independent one, and we showed that the compositional question usually posed as a Markov assumption is in fact an identity plus a transportability hypothesis. On five models we find survival to be strongly role-dependent, transportability to fail by a factor of four on one frontier model while holding on another, and defence efficacy to be unidentifiable without a matched no-defence baseline. We also report a significant result of our own that did not survive a protocol fix, together with the fix, because in a measurement study the protocol is part of the result. Code, configurations and raw model outputs for every cell are released with the paper, including the per-trial artefacts needed to reproduce the ablation of Section 5.3.